of Machine Learning Research, 1921–1931. Online: PMLR. URL http://proceedings.mlr.press/v108/nguyen20a.html.

Puterman, M. L. 2014. *Markov decision processes: discrete stochastic dynamic programming*. John Wiley & Sons.

Rebeschini, P. 2019. Oxford Algorithmic Foundations of Learning, Lecture Notes: Sub-Gaussian Concentration Inequalities. Bounds in Probability. URL: http://www.stats.ox.ac.uk/~rebeschi/teaching/AFoL/20/material/lecture06.pdf. Last visited on Sep. 14, 2020.

Rowland, M.; Bellemare, M. G.; Dabney, W.; Munos, R.; and Teh, Y. W. 2018. An Analysis of Categorical Distributional Reinforcement Learning. In *AISTATS*, volume 84 of *Proceedings of Machine Learning Research*, 29–37. PMLR.

Rowland, M.; Dadashi, R.; Kumar, S.; Munos, R.; Bellemare, M. G.; and Dabney, W. 2019. Statistics and Samples in Distributional Reinforcement Learning. In *ICML*, volume 97 of *Proceedings of Machine Learning Research*, 5528–5536. PMLR.

Rummery, G. A.; and Niranjan, M. 1994. On-Line Q-Learning Using Connectionist Systems.

Schaul, T.; Quan, J.; Antonoglou, I.; and Silver, D. 2016. Prioritized Experience Replay. In *ICLR (Poster)*.

Smola, A. J.; Gretton, A.; Song, L.; and Schölkopf, B. 2007. A Hilbert Space Embedding for Distributions. In *ALT*, volume 4754 of *Lecture Notes in Computer Science*, 13–31. Springer.

Sriperumbudur, B. K.; Fukumizu, K.; Gretton, A.; Lanckriet, G. R. G.; and Schölkopf, B. 2009. Kernel Choice and Classifiability for RKHS Embeddings of Probability Distributions. In *NIPS*, 1750–1758. Curran Associates, Inc.

Sutton, R. S. 1988. Learning to Predict by the Methods of Temporal Differences. *Mach. Learn.* 3: 9–44.

Sutton, R. S.; Barto, A. G.; et al. 1998. *Introduction to reinforcement learning*, volume 135. MIT press Cambridge.

Székely, G. J. 2003. E-statistics: The energy of statistical samples. *Bowling Green State University, Department of Mathematics and Statistics Technical Report* 3(05): 1–18.

van Hasselt, H.; Guez, A.; and Silver, D. 2016. Deep Reinforcement Learning with Double Q-Learning. In *AAAI*, 2094–2100. AAAI Press.

Wang, Z.; Schaul, T.; Hessel, M.; van Hasselt, H.; Lanctot, M.; and de Freitas, N. 2016. Dueling Network Architectures for Deep Reinforcement Learning. In *ICML*, volume 48 of *JMLR Workshop and Conference Proceedings*, 1995–2003. JMLR.org.

Watkins, C. J. C. H.; and Dayan, P. 1992. Q-learning. In *Machine Learning*, 279–292.

Welling, M. 2009. Herding dynamical weights to learn. In *ICML*, volume 382 of *ACM International Conference Proceeding Series*, 1121–1128. ACM.

Yang, D.; Zhao, L.; Lin, Z.; Qin, T.; Bian, J.; and Liu, T.-Y. 2019. Fully Parameterized Quantile Function for Distributional Reinforcement Learning. In *Advances in Neural Information Processing Systems*, 6190–6199.

Appendix A. Proofs

A.1. Proof of Proposition 1

Proof. If MMD is a metric in $\tilde{\mathcal{P}}(\mathcal{X})$. Then, it is obvious to see that $\text{MMD}_\infty(\mu,\nu) \geq 0, \forall \mu,\nu$ and that $\text{MMD}_\infty(\mu,\nu) = 0$ implies $\mu = \nu$. We now prove that $\texttt{MMD}_\infty$ satisfies the triangle inequality. Indeed, for any $\mu,\nu,\eta \in \tilde{\mathcal{P}}(\mathcal{X})^{\mathcal{S}\times\mathcal{A}}$, we have

$$\begin{aligned}
\text{MMD}_\infty(\mu,\nu) &= \sup_{(s,a)\in\mathcal{S}\times\mathcal{A}} \text{MMD}(\mu(s,a),\nu(s,a)) \\
&\overset{(a)}{\leq} \sup_{(s,a)\in\mathcal{S}\times\mathcal{A}} \Big\{ \text{MMD}(\mu(s,a),\eta(s,a)) + \text{MMD}(\eta(s,a),\nu(s,a)) \Big\} \\
&\overset{(b)}{\leq} \sup_{(s,a)\in\mathcal{S}\times\mathcal{A}} \text{MMD}(\mu(s,a),\eta(s,a)) + \sup_{(s,a)\in\mathcal{S}\times\mathcal{A}} \text{MMD}(\eta(s,a),\nu(s,a)) \\
&= \text{MMD}_\infty(\mu,\eta) + \text{MMD}_\infty(\eta,\nu),
\end{aligned}$$

where (a) follows from the triangle inequality for MMD and (b) follows from that $\sup(A+B) \leq \sup A + \sup B$ for any two sets A and B where $A + B := \{a + b : a \in A, b \in B\}$.

□

A.2. Proof of Theorem 1

We first present a relevant result for the proof.

Lemma 2. *Let ϕ be the feature vector of k, i.e., $k(x,y) = \phi(x)^T\phi(y)$. Then, for any $\mu,\nu \in \mathcal{P}(\mathcal{X})$, we have*

$$MMD(\mu;\nu;k) = \|u - v\|_2$$

where $u = \mathbb{E}_{x\sim\mu}\phi(x)$ and $v = \mathbb{E}_{y\sim\nu}\phi(y)$.

Proof. Let $X, X' \overset{\text{i.i.d.}}{\sim} \mu$, $Y, Y' \overset{\text{i.i.d.}}{\sim} \nu$, and X, X', Y, Y' are mutually independent. We have

$$\begin{aligned}
\text{MMD}^2(\mu;\nu;k) &= \mathbb{E}\left[k(X,X')\right] + \mathbb{E}\left[k(Y,Y')\right] - 2\left[\mathbb{E}k(X,Y)\right] \\
&= \mathbb{E}\left[\phi(X)^T\phi(X')\right] + \mathbb{E}\left[\phi(Y)^T\phi(Y')\right] - 2\mathbb{E}\left[\phi(X)^T\phi(Y)\right] \\
&= u^Tu + v^Tv - 2u^Tv \\
&= \|u - v\|^2.
\end{aligned}$$

□

Proof of Theorem 1. We prove only Theorem 1.3, as the proofs of Theorem 1.1 and 1.2 can be preferred to in (Fukumizu et al. 2007; Gretton et al. 2012) and (Székely 2003, c.f. Proposition 2), respectively. For some $\sigma > 0$, let $k(x,y) = \exp(xy/\sigma^2)$. Let $\phi(x) = [a_0(x), a_1(x), ..., a_n(x), ...]$ where $a_n(x) = \frac{1}{\sqrt{n!}}\frac{x^n}{\sigma^n}$. The Taylor expansion of k yields $k(x,y) = \phi(x)^T\phi(y)$. It follows from Lemma 2 that

$$\begin{aligned}
\text{MMD}^2(\mu,\nu;k) &= \|\mathbb{E}\phi(X) - \mathbb{E}\phi(Y)\|^2 \\
&= \sum_{n=0}^{\infty} \frac{1}{\sigma^{2n}n!}\left(\mathbb{E}[X^n] - \mathbb{E}[Y^n]\right)^2,
\end{aligned}$$

for any $\mu,\nu \in \mathcal{P}(\mathcal{X})$ where $X \sim \mu$ and $Y \sim \nu$. It is easy to see that $\text{MMD}(\mu,\nu;k) \geq 0$ and it satisfies the triangle inequality. We only need to prove that for any $\mu,\nu \in \mathcal{P}_*(\mathcal{X})$, if $\text{MMD}(\mu,\nu;k) = 0$, then $\mu = \nu$. Indeed, assume $\text{MMD}(\mu,\nu;k) = 0$, then μ and ν have equal moments of all orders. Note that a distribution μ is uniquely determined by its characteristic function $g_\mu(t) = \mathbb{E}\left[\exp(itX)\right], \forall t$. Let $m_n(\mu) = \mathbb{E}[X^n]$ be the n-th moment of μ. Taylor expansion of g_μ yields

$$g_\mu(t) = \sum_{n=0}^{\infty} \frac{i^n t^n m_n(\mu)}{n!},$$

a power series which is valid only within its radius of convergence. The radius of convergence of this power series is

$$r = \frac{1}{\limsup_{n\to\infty}\left|\frac{m_n(\mu)}{n!}\right|^{1/n}}.$$

Since $\mu \in \mathcal{P}_*(\mathcal{X})$, we have

$$\limsup_{n\to\infty} \frac{|m_n(\mu)|^{1/n}}{n} = 0.$$

Using Stirling's formular, this indicates that $\limsup_{n\to\infty} \left|\frac{m_n(\mu)}{n!}\right|^{1/n} = 0$, or $r = \infty$. Hence, the set of all moments of a distribution on $\mathcal{P}_*(\mathcal{X})$ uniquely determines the distribution. This concludes our proof. $\square$

A.3. Proof of Theorem 2

Proof of Theorem 2.1

Lemma 3. *Let $(\mu_i)_{i\in I}$ and $(\nu_i)_{i\in I}$ be two sets of Borel probability measures in $\mathcal{X}$ over some indices I. Let p be any distribution induced over I, then we have*

$$MMD^2\left(\sum_i p_i\mu_i, \sum_i p_i\nu_i\right) \leq \sum_i p_i MMD^2(\mu_i,\nu_i)$$

Proof. Denoting $g_i = \psi_{\mu_i} - \psi_{\nu_i}, \forall i$, we have

$$\begin{aligned}
\mathrm{MMD}^2\left(\sum_i p_i\mu_i, \sum_i p_i\nu_i\right) &= \left\|\psi_{\sum_i p_i\mu_i} - \psi_{\sum_i p_i\nu_i}\right\|_{\mathcal{F}}^2 \\
&\overset{(a)}{=} \left\|\sum_i p_i(\psi_{\mu_i} - \psi_{\nu_i})\right\|_{\mathcal{F}}^2 \\
&= \sum_i \langle p_i g_i, p_i g_i\rangle_{\mathcal{F}} + 2\sum_{i\neq j}\langle p_i g_i, p_j g_j\rangle_{\mathcal{F}} \\
&= \sum_i p_i^2\langle g_i, g_i\rangle_{\mathcal{F}} + 2\sum_{i\neq j} p_i p_j\langle g_i, g_j\rangle_{\mathcal{F}} \\
&\overset{(b)}{\leq} \sum_i p_i^2\|g_i\|_{\mathcal{F}}^2 + 2\sum_{i\neq j} p_i p_j\|g_i\|_{\mathcal{F}}\|g_j\|_{\mathcal{F}} \\
&= \left(\sum_i \sqrt{p_i}\sqrt{p_i}\|g_i\|_{\mathcal{F}}\right)^2 \\
&\overset{(c)}{\leq} (\sum_i p_i)\sum_i p_i\,\|g_i\|_{\mathcal{F}}^2 \\
&= \sum_i p_i\,\|\psi_{\mu_i} - \psi_{\nu_i}\|_{\mathcal{F}}^2 = \sum_i p_i \mathrm{MMD}^2(\mu_i,\nu_i),
\end{aligned}$$

where (a) follows from that the Bochner integral is linear (w.r.t. the probability measure argument), i.e., $\psi_{\sum_i p_i\mu_i} = \sum_i p_i\psi_{\mu_i}$, and both inequalities (b) and (c) follow from Cauchy-Schwartz inequality. $\square$

Lemma 4. *If k is shift invariant and scale sensitive with order $\alpha > 0$. Then for any $\mu,\nu \in \mathcal{P}(\mathcal{X})$ and any $r,\gamma \in \mathbb{R}$, we have*

$$MMD^2\Big((f_{r,\gamma})_{\#}\mu, (f_{r,\gamma})_{\#}\nu; k\Big) = |\gamma|^\alpha MMD^2(\mu,\nu;k),$$

where $f_{r,\gamma}(z) := r + \gamma z$ and $\#$ denotes pushforward operator.